

4.4 Conclusion

AI tools don't replace manual refactoring, but they are invaluable aids that can relieve you of work and support you, for example, in not only analyzing the source code but also implementing a refactoring process. A key advantage of these AI tools is their ability to see patterns in code that can be difficult for the human eye to recognize. They analyze the source code in a depth and speed that can't be achieved by a developer alone.

Not only do these tools provide concrete suggestions for improvements, they often also explain the background to their recommendations. This al-



explanations and tips, AI tools promote a deeper understanding of the underlying problems and the proposed solutions.

Another significant advantage is the consistency guaranteed by the use of AI tools. While human developers can make mistakes due to fatigue or time pressure, AI algorithms work tirelessly and without subjective influences. This ensures that refactoring is carried out consistently and according to best practices.

However, it's important to note that AI tools need to be monitored, as they can occasionally overshoot the mark and introduce errors through hallucinations. These hallucinations can lead to the AI tool suggesting changes that don't correspond to the actual requirements or the desired behavior of the system. It's therefore crucial that human developers carefully review and validate the changes suggested by AI.

In summary, AI-supported tools are a valuable addition to manual refactoring. They improve the efficiency and quality of the refactoring process and allow developers to focus on more creative and challenging tasks. Despite these benefits, it's important to emphasize that human expertise and criti-

cal thinking are still essential to fully understand and consider the context and specific requirements of a project.